

# ECE 447

## Lesson 09

### Amplitude

### Modulation, Part 2



UNITED STATES  
AIR FORCE  
ACADEMY

# SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
  - **HW1.** was due last lesson (Lesson 8, 26 Aug)
  - **Lab 2 Assignment.** The assignment associated with Lab 2 is due Lesson 10 (Tue, 1 Sep) by 2359 via Gradescope upload.
  - **HW 2.** Due Lesson 11

## DOUBLE-SIDEBAND, LARGE-CARRIER (DSB-LC)

- Add a signal at the carrier to the DSB-SC signal from last lesson
- Now we have the conventional AM used in broadcast signals (AM radio, Airband)
- Time domain:

$$\varphi_{AM}(t) = A\cos(\omega_c t) + m(t)\cos(\omega_c t) = [A + m(t)]\cos(2\pi f_c t)$$

- Frequency domain:

$$\varphi_{AM}(t) \Leftrightarrow \frac{1}{2}[M(f + f_c) + M(f - f_c)] + \frac{A}{2}[\delta(f + f_c) + \delta(f - f_c)]$$

# AM: 215 vs. 447

## ECE 215:

- Sinusoidal messages (tones) only:

$$A_c \cos(2\pi \cdot f_c \cdot t) \cdot [A_m \cos(2\pi \cdot f_m \cdot t) + B] = \frac{A_c A_m}{2} \cos[2\pi \cdot (f_c - f_m) \cdot t] + \frac{A_c A_m}{2} \cos[2\pi \cdot (f_c + f_m) \cdot t] + A_c B \cos(2\pi \cdot f_c \cdot t)$$

- Modulation Index:  $\alpha = \frac{A_m}{B} = \frac{2 \cdot \text{Amp}_{SB}}{\text{Amp}_{Carrier}}$
- Efficiency:  $\eta = \frac{\alpha^2}{\alpha^2 + 2}$

## ECE 447:

- Any arbitrary message:

$$\varphi_{AM}(t) = A \cos(2\pi f_c t) + m(t) \cos(2\pi f_c t)$$

- Modulation Index:  $\mu = \frac{m_p}{A}$ , where  $m_p$  is the max value of  $m(t)$
- Efficiency:  $\eta = \frac{m(t)^2}{A^2 + m(t)^2}$  in general, and for tone modulation  $\eta = \frac{\mu^2}{\mu^2 + 2}$

# EXAMPLE PROBLEM

4.3-5 Consider an AM signal with  $m(t)$  shown in Fig. P4.3-5 with  $f_c = 10$  kHz and  $\mu = 2$ .

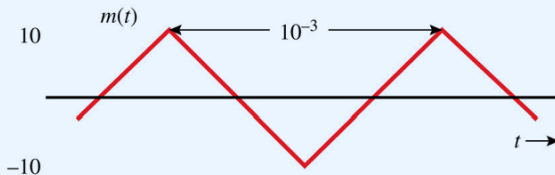
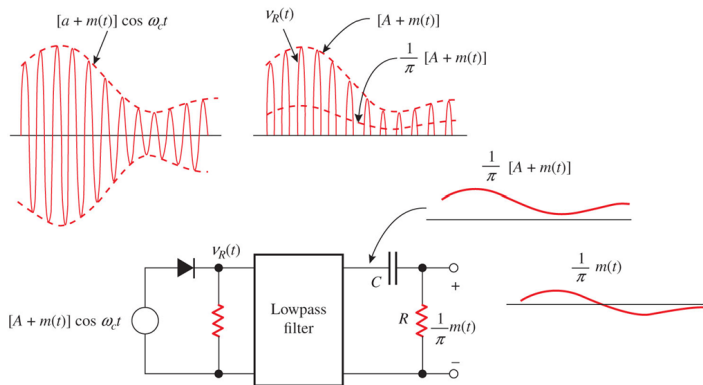


FIGURE P4.3-5 [Enlarge image](#)

- Find the amplitude and power of the carrier.
- Find the sideband power and the power efficiency  $\eta$ .

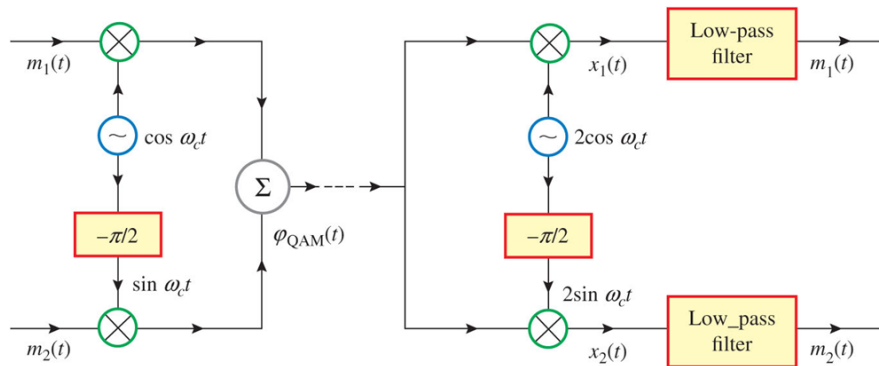
# ENVELOPE DETECTOR DEMODULATION



# BW-EFFICIENT MODULATIONS

- DSB modulations use twice as much BW as necessary - the complete message info is contained in both the USB and LSB
- **One solution:** remove all or most of one of the sidebands!
  - **Single sideband, suppressed-carrier (SSB-SC):** used in two-way radio (HF shortwave) and HAM radio
  - **Vestigial sideband, suppressed-carrier (VSB-SC):** used in old analog TV video signals and modern ATSC Digital TV Standard to get slightly more BW than straight SSB
- **Quadrature Amplitude Modulation (QAM)**
  - Used in most modern wireless communication systems: Wi-Fi (all recent variants), 4G/5G cellular systems, satellite TV and radio, broadband internet
  - You will see this in Lab 3 simulations, and we will cover QAM in much more detail in the future...

# QAM



- $\varphi_{QAM}(t) = m_1(t)\cos(\omega_c t) + m_2(t)\sin(\omega_c t)$